KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5501 | 5 Credits: 4 | Program Core | Source-grounded

Linear Integrated Circuits

 To provide an insight on the fundamental concepts of operational amplifiers and their

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5501 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5501.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electronics and Computer Engineering
Course code	5501
Course title	Linear Integrated Circuits
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester:60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide an insight on the fundamental concepts of operational amplifiers and their
- applications.
- To give a functional description of various linear integrated circuits and discuss on
- various circuits which act as the bridge between analog and digital domains.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding electrical parameters of operational amplifier.	See official syllabus
CO2	Develop circuits using operational amplifier. 20 Applying	See official syllabus
CO3	Implement circuits using timer IC and PLL 12 Applying Explain the operation of various fixed and	See official syllabus
CO4	14 Understanding variable voltage regulators, ADC and DAC	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	operational amplifier.	4	As specified
Module 2	Develop circuits using operational amplifier.	5	As specified
Module 3	Implement circuits using timer IC and PLL	4	As specified
Module 4	ADC and DAC	3	As specified

MODULE 1

operational amplifier.

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: operational amplifier.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

4 Understanding amplifier Explain the block diagram of operational

4 Understanding amplifier Explain the block diagram of operational is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding amplifier Explain the block diagram of operational using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding amplifier explain the block diagram of operational should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding amplifier Explain the block diagram of operational means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding amplifier Explain the block diagram of operational □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding amplifier

2 Understanding amplifier is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding amplifier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding amplifier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding amplifier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding amplifier [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain operational amplifier parameters

Explain operational amplifier parameters is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain operational amplifier parameters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain operational amplifier parameters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain operational amplifier parameters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain operational amplifier parameters [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower

Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate thebasic op-amp feedback circuits 3 understanding fundamentals of operational amplifier differential amplifier: bjt differential amplifier - types -large and small signal operation- input resistance - voltage gain - cmrr - non-ideal characteristics operational amplifier : op-amp symbol - package types - block diagram - dc and ac op-amp parameters - equivalent circuit - voltage transfer curve - open loop op-amp configurations - op-amp with negative feedback - properties of practical op-amp - inverting amplifier - non-inverting amplifier - voltage follower should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
operational amplifier.
            Illustrate the operation of differential
  M1.01                                                        4
Understanding
            amplifier
            Explain the block diagram of operational
  M1.02                                                        2
Understanding
            amplifier
  M1.03     Explain operational amplifier parameters           3
Understanding

  M1.04     Illustrate thebasic op-amp feedback circuits       3
Understanding

Contents:
Fundamentals of operational amplifier
Differential amplifier: BJT differential amplifier - types -Large and small signal
operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics
Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC
op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp
configurations - Op-amp with negative feedback - Properties of practical op-amp -
Inverting amplifier - Non-inverting amplifier - Voltage follower
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Develop circuits using operational amplifier.

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop circuits using operational amplifier.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Develop various amplifiers using op-amp

Develop various amplifiers using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop various amplifiers using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop various amplifiers using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop various amplifiers using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Develop various amplifiers using op-amp [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Developwave-shaping circuits using op-amp

Developwave-shaping circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Developwave-shaping circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, developwave-shaping circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Developwave-shaping circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Developwave-shaping circuits using op-amp [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop oscillator circuits using op-amp

Develop oscillator circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop oscillator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop oscillator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop oscillator circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Develop oscillator circuits using op-amp [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate comparator circuits using op-amp

Illustrate comparator circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate comparator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate comparator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate comparator circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate comparator circuits using op-amp □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters.

Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, developactive filters using op-amp 4 applying applications of operational amplifier summing amplifier - difference amplifier - scaling and averaging amplifiers - adder - subtractor - instrumentation amplifier - voltage to current converter, current to voltage converter - integrator - differentiator - precision rectifiers - peak detector - clipper and clamper - log and antilog amplifier - phase shift and wien bridge oscillators - astable and mono stable multi vibrators - triangular and saw tooth wave generators- comparators - zero crossing detector - schmitt trigger - first order low-pass, high-pass and band-pass butterworth filters. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass

Butterworth filters. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Develop circuits using operational amplifier.

  M2.01     Develop various amplifiers using op-amp            4
Applying

  M2.02     Developwave-shaping circuits using op-amp          4
Applying

  M2.03     Develop oscillator circuits using op-amp           4
Applying

  M2.04     Illustrate comparator circuits using op-amp        4
Applying

  M2.05     Developactive filters using op-amp                 4
Applying
            Series Test I                                      1
Contents:
Applications of operational amplifier
Summing amplifier - difference amplifier - Scaling and averaging amplifiers -
adder -
subtractor - Instrumentation amplifier - Voltage to current converter, Current to
voltage
converter - Integrator - Differentiator - Precision rectifiers - Peak detector -
Clipper and
Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators -
Astable
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Implement circuits using timer IC and PLL

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Implement circuits using timer IC and PLL
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding timer

3 Understanding timer is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding timer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding timer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding timer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding timer [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565

Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop circuits using 555 timer ic 3 applying explain the functional block diagram of 565 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding PLL

3 Understanding PLL is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding PLL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding pll should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding PLL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding PLL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators,

Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate applications ofpll 3 understanding timer & pll 555 timer ic- functional block diagram - features – a stable and mono stable multi vibrator using 555 - phase locked loop - general block diagram of pll - operation - lock range, capture range and pull-in time - functional block diagram of pll ic 565 - applications of pll for am & fm detection and frequency multiplication, frequency division, frequency synthesizing explain the operation of various fixed and variable voltage regulators, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Implement circuits using timer IC and PLL

             Explain the functional block diagram of 555
  M3.01                                                           3
Understanding
             timer
  M3.02      Develop circuits using 555 timer IC                  3
Applying
             Explain the functional block diagram of 565
  M3.03                                                           3
Understanding
             PLL
  M3.04      Illustrate applications ofPLL                        3
Understanding

Contents:
Timer & PLL
555 Timer IC- Functional block diagram - features – a stable and mono stable
multi
vibrator using 555 -
Phase Locked Loop - General block diagram of PLL - Operation - Lock range,
capture
range and pull-in time - Functional block diagram of PLL IC 565 - Applications of
PLL
for AM & FM detection and Frequency multiplication, Frequency division, Frequency
synthesizing
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

ADC and DAC

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: ADC and DAC
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to

Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain fixed and variable voltage regulators 4 understanding compare and contrast various analog to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Understanding digital converters Compare and contrast various digital to

5 Understanding digital converters Compare and contrast various digital to is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding digital converters Compare and contrast various digital to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding digital converters compare and contrast various digital to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding digital converters Compare and contrast various digital to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Understanding digital converters Compare and contrast various digital to □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits

5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3

https://nptel.ac.in/courses/108/108/108108111/ 4
https://inst.eecs.berkeley.edu/~ee140/fa19/ 5
https://freevideolectures.com/course/2915/linear-integrated-circuits is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding analog converters ic voltage regulators and data converters monolithic voltage regulators: fixed voltage regulators, 78xx and 79xx series - adjustable voltage regulators ic lm723 - low voltage and high voltage configurations, current boosting, current limiting, short circuit and fold-backprotection- optocoupler - principle of operation. data converters:d/a converter - specifications - weighted resistor type, r-2r ladder type. a/d converters: specifications - classification - flash type - counter ramp type - successive approximation type - dual slope type - sample-and-hold circuits. text / reference: t/r book title/author roy d. c. and s. b. jain, linear integrated circuits, new age international, t1 3/e, 2010 sergio franco, design with operational amplifiers and analog integrated t2 circuits, 3rd edition, tata mcgraw-hill, 2008 gayakwad r. a., op-amps and linear integrated circuits, prentice hall, r1 4/e, 2010 r2 botkar k. r., integrated circuits, 10/e, khanna publishers, 2010 salivahanans. ,v. s. k. bhaaskaran, linear integrated circuits, tata r3 mcgraw hill, 2008 david a. bell, operational amplifiers & linear ics, oxford university r4 press, 2nd edition,2010 online resources: sl.no website link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
ADC and DAC

  M4.01      Explain fixed and variable voltage regulators        4
Understanding
             Compare and contrast various analog to
  M4.02                                                           5
Understanding
             digital converters
             Compare and contrast various digital to
  M4.03                                                           5
Understanding
             analog converters
             Series Test II                                       1
Contents:
IC Voltage regulators and Data Converters
Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series -
Adjustable voltage regulators IC LM723 - Low voltage and high voltage
configurations,
Current boosting, Current limiting, Short circuit and Fold-backprotection-
optocoupler -
principle of operation.
Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R
Ladder
type.
 A/D Converters: Specifications - Classification - Flash type - Counter ramp type
-
 Successive approximation type - Dual slope type - Sample-and-hold circuits.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 4 Understanding amplifier Explain the block diagram of operational. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding amplifier Explain the block diagram of operational*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding amplifier. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding amplifier*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain operational amplifier parameters. (3 marks)

Answer: Begin with the standard definition or identity of *Explain operational amplifier parameters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate thebasic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-*

amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Develop various amplifiers using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Develop various amplifiers using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Developwave-shaping circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Developwave-shaping circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Develop oscillator circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Develop oscillator circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate comparator circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate comparator circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 3 Understanding timer. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding timer*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565. (3 marks)

Answer: Begin with the standard definition or identity of *Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding PLL. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding PLL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators,. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate applications ofPLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features – a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to. (3 marks)

Answer: Begin with the standard definition or identity of *Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding digital converters Compare and contrast various digital to. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding digital converters Compare and contrast various digital to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd

edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Understanding amplifier Explain the block diagram of operational**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Develop various amplifiers using op-amp**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Understanding timer**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://nptel.ac.in/courses/117107094/ https://nptel.ac.in/courses/117101106/
- https://nptel.ac.in/courses/108/108/108108111/ https://inst.eecs.berkeley.edu/~ee140/fa19/
- https://freevideolectures.com/course/2915/linear-integrated-circuits

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